



CANDIDATE – PLEASE NOTE!

PRINT your name on the line below and return this booklet with your answer sheet. Failure to do so may result in disqualification.

TEST CODE **01238010**

MAY/JUNE 2018

FORM TP 2018096

CARIBBEAN EXAMINATIONS COUNCIL
CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION

PHYSICS

Paper 01 – General Proficiency

1 hour 15 minutes

06 JUNE 2018 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This test consists of 60 items. You will have 1 hour and 15 minutes to answer them.
2. In addition to this test booklet, you should have an answer sheet.
3. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
4. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

The SI unit of length is the

- (A) metre
- (B) second
- (C) newton
- (D) kilogram

Sample Answer



The best answer to this item is “metre”, so (A) has been shaded.

5. If you want to change your answer, erase it completely before you fill in your new choice.
6. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to that item later.
7. Figures are not necessarily drawn to scale.
8. You may do any rough work in this booklet.
9. You may use a silent, non-programmable calculator to answer items.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

1. Errors due to parallax can be minimized by

- (A) taking more than one reading
- (B) placing the eye at right angles to the mark being read
- (C) taking readings from different angles
- (D) taking an average of two readings using two separate scales

2. Which of the following measuring instruments is MOST suitable for accurately measuring the volume of 40.2 cm^3 of a liquid?

- (A) Burette
- (B) Pipette
- (C) Beaker
- (D) Measuring cylinder

3. Which of the following changes can be caused by a force acting on a body?

- I. Shape
- II. Motion
- III. Density

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

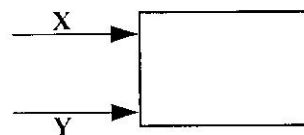
4. The mass of an astronaut is 70 kg when standing on the moon. When he returns to Earth, his approximate weight will be

- (A) 70 kg
- (B) 70 N
- (C) 420 kg
- (D) 700 N

5. The moment of a force is defined as the

- (A) moment in time when a force is first applied to a body
- (B) length of time for which a force is applied to a body
- (C) product of the force and its perpendicular distance from the turning point to the force
- (D) ratio of the force and its perpendicular distance from the turning point

Item 6 refers to the following diagram which shows two forces, X and Y, applied to an object.



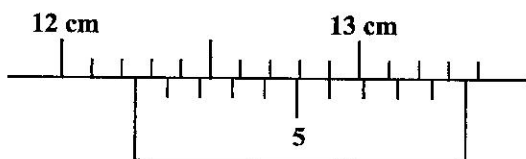
6. What should be the magnitude and direction of a third force which will cause the object to remain stationary?

- (A) $X - Y$ to the left
- (B) $X + Y$ to the left
- (C) $X - Y$ to the right
- (D) $X + Y$ to the right

7. When a force, F , is applied to a spring of original length, L , the new length becomes $L + x$. What would be the length of the spring if a force of $\frac{F}{2}$ is applied instead?

- (A) $L + 2x$
- (B) $2(L + x)$
- (C) $L + x$
- (D) $L + \frac{x}{2}$

Item 8 refers to the following diagram of a vernier caliper.



8. Which of the following readings is shown on the scale above?

(A) 12.06 cm
(B) 12.26 cm
(C) 12.25 cm
(D) 12.29 cm

9. Which of the following measures may be classified as scalar quantities?

I. Time
II. Speed
III. Displacement

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

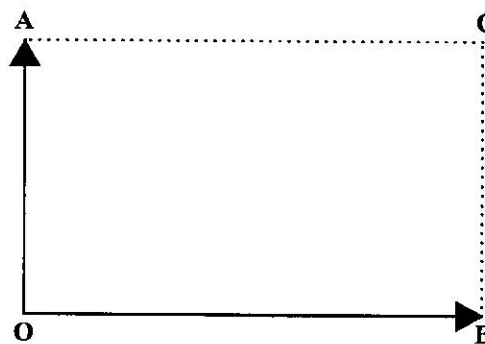
10. Pressure in a liquid can be calculated using the formula

$$P = \rho gh.$$

Which of the following sets of units will result in the SI unit of pressure?

	ρ	g	h
(A)	g cm^{-3}	m s^{-2}	mm
(B)	g cm^{-3}	N kg^{-1}	m
(C)	kg m^{-3}	cm s^{-2}	cm
(D)	kg m^{-3}	N kg^{-1}	m

Item 11 refers to the following diagram which shows two vectors of magnitudes a and b represented respectively by $\overline{OA}$ and $\overline{OB}$.



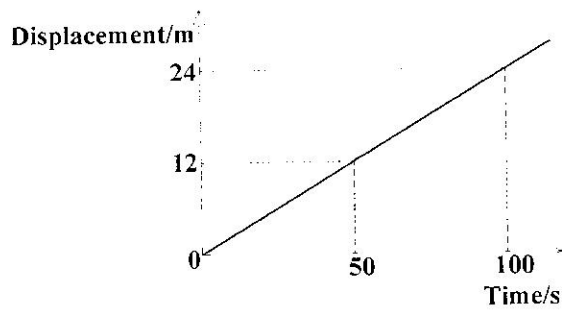
11. The vectors act at point O and are directed perpendicular to each other. Which of the following pairs represents BOTH the magnitude and direction of their resultant?

	Magnitude	Direction
(A)	$a^2 + b^2$	$\overline{OC}$
(B)	$a + b$	$\overline{CO}$
(C)	$\sqrt{(a^2 + b^2)}$	$\overline{CO}$
(D)	$\sqrt{(a^2 + b^2)}$	$\overline{OC}$

12. The acceleration due to gravity (g) is to be determined by measuring the time (t) taken for a small steel ball to fall through a specific height. Which of the following activities is UNNECESSARY?

(A) Find the mass of the ball.
(B) Allow the ball to drop and start the stopwatch at the same instant.
(C) Measure the height through which the ball falls.
(D) Repeat the time measurements and take the average.

Item 13 refers to the following graph which shows how the displacement of a runner from a starting line varies with time.



13. From the graph it can be deduced that the runner is

(A) not moving
(B) going at a steady speed
(C) going faster and faster
(D) going slower and slower

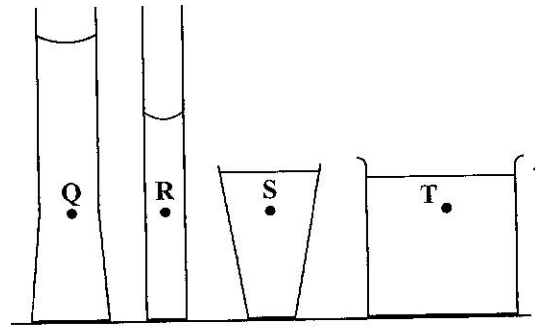
14. The time period of a simple pendulum oscillating with a small amplitude depends on the

(A) mass of the pendulum bob
(B) amplitude of the oscillation
(C) length of the pendulum
(D) force with which the pendulum is set into motion

15. An object is removed from the ground and placed on a shelf. Which of the object's properties is expected to increase?

(A) Mass
(B) Volume
(C) Potential energy
(D) Kinetic energy

Item 16 refers to the following diagrams of four vessels, filled with water.



16. The pressure due to the water is the GREATEST at

(A) Q
(B) R
(C) S
(D) T

17. A bubble of gas rises to the surface of a soft drink. This is because the

(A) density of the gas is greater than the density of the drink
(B) upthrust on the bubble is greater than the weight of the bubble
(C) upthrust on the bubble is greater than the weight of water it displaces
(D) weight of the water displaced by the bubble is less than the weight of the bubble

18. Which of the following statements about the pressure law are true?

- I. The ratio of pressure to Kelvin temperature is constant.
- II. Volume is constant.
- III. Pressure is constant.

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

19. Which of the following statements concerning the radiation of heat is/are true?

- I. Radiation can only take place in a material medium.
- II. A good absorber of radiation is also a good emitter of radiation.
- III. Dark, dull surfaces are better emitters than shiny ones.

- (A) III only
- (B) I and II only
- (C) I and III only
- (D) II and III only

20. An electric kettle full of water is plugged into the mains. The process by which heat travels through the water is known as

- (A) radiation
- (B) convection
- (C) evaporation
- (D) electrification

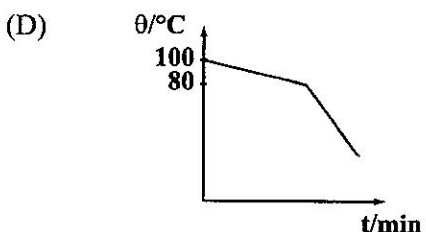
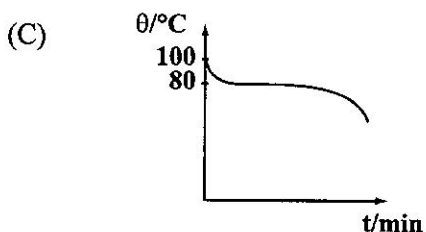
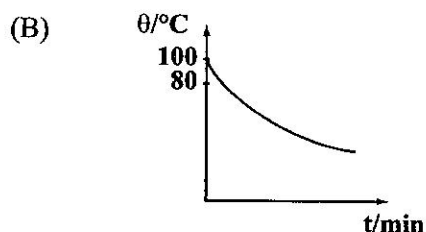
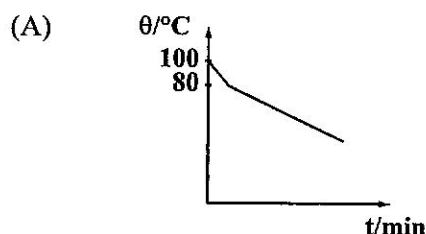
21. The energy required to change the state of a substance was determined to be E_H . If the mass of the substance is DOUBLED, the value of E_H will

- (A) be halved
- (B) be doubled
- (C) be quadrupled
- (D) remain constant

22. Which of the following statements about evaporation is FALSE?

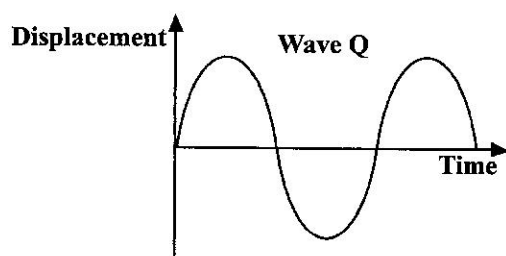
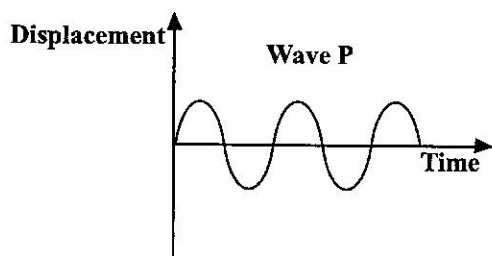
- (A) Evaporation occurs at room temperature only.
- (B) Evaporation requires heat energy and causes cooling.
- (C) Evaporation occurs only at the surface.
- (D) In evaporation the faster molecules escape the liquid.

23. Some molten naphthalene at 100°C is allowed to cool to room temperature. If naphthalene has a melting point of 80°C , which of the following graphs BEST represents the cooling curve of naphthalene?



24. The specific latent heat of vaporization of water is the energy required to change 1 kg of water at
- (A) 0 °C to ice at 0 °C
 - (B) 0 °C to steam at 100 °C
 - (C) 100 °C to steam at 100 °C
 - (D) 99.9 °C to steam at 100.1 °C
25. A metal of mass m requires energy, E , to raise its temperature from T_1 to T_2 . The specific heat capacity of the metal will be given by
- (A) $\frac{E}{mT_2}$
 - (B) $\frac{Em}{(T_1 - T_2)}$
 - (C) $\frac{E}{m(T_1 - T_2)}$
 - (D) $\frac{E}{m(T_2 - T_1)}$
26. When heating a substance, a point is reached where no further change in temperature is observed. The MOST likely reason for this is that the
- (A) heater is supplying less heat
 - (B) substance is changing state
 - (C) substance needs to be stirred less frequently
 - (D) substance is losing heat to the atmosphere
27. Which of the following phenomena are exhibited by sound waves?
- I. Refraction
 - II. Diffraction
 - III. Interference
- (A) I and II only
 - (B) I and III only
 - (C) II and III only
 - (D) I, II and III
28. Whenever there is complete destructive interference between two coherent wave trains, the waves must be
- (A) in phase
 - (B) out of phase by half of a wavelength
 - (C) out of phase by one wavelength
 - (D) out of phase by a quarter of a wavelength

Item 29 refers to the following graphs (with axes having the same scales) of two sound waves, P and Q.

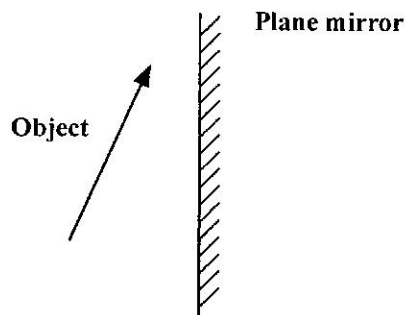


29. Which of the following statements is true?
- (A) P is louder than Q but Q has a higher pitch.
 - (B) P is louder than Q and also has a higher pitch than Q.
 - (C) Q is louder than P but P has a higher pitch.
 - (D) Q is louder than P and also has a higher pitch than P.

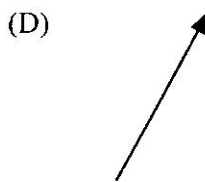
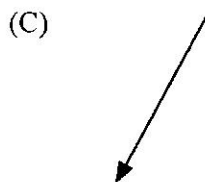
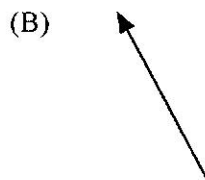
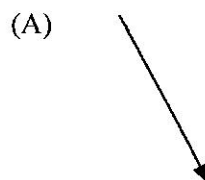
30. Which of the following equations expresses the correct relationship between the wavelength, λ , speed, v , and frequency, f , of a wave?

- (A) $f = \frac{v}{\lambda}$
- (B) $f = \frac{\lambda}{v}$
- (C) $\lambda = fv$
- (D) $\lambda = \frac{f}{v}$

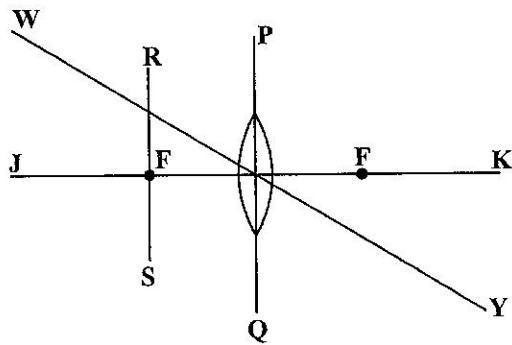
Item 31 refers to the following diagram which shows an object in front of a plane mirror.



31. Which of the following options BEST represents the image produced by the plane mirror?



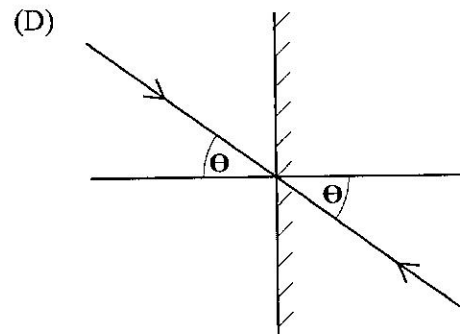
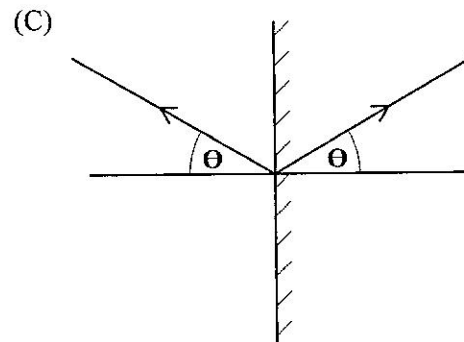
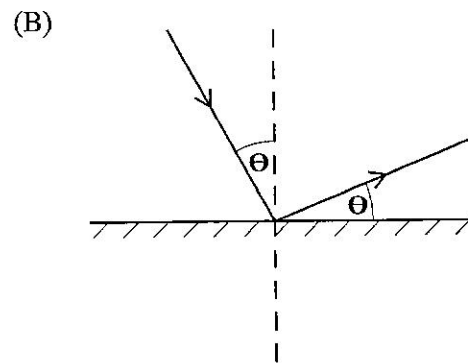
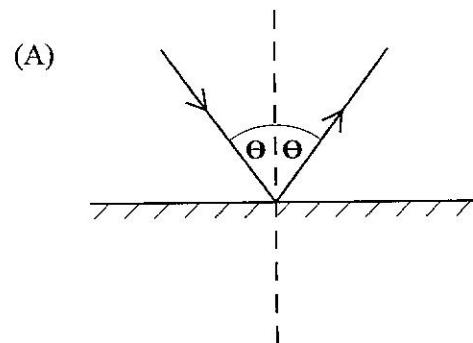
Item 32 refers the following diagram.



32. Which line lies along the principal axis of the converging lens?

- (A) JK
- (B) PQ
- (C) RS
- (D) WY

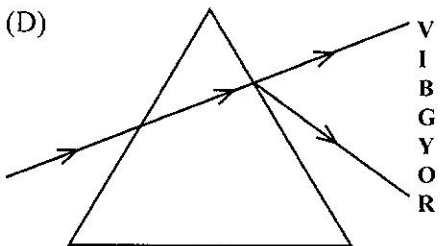
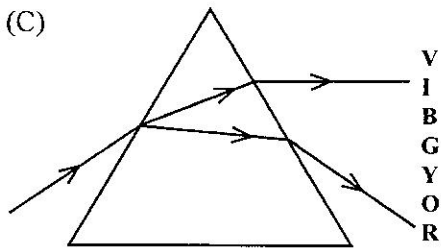
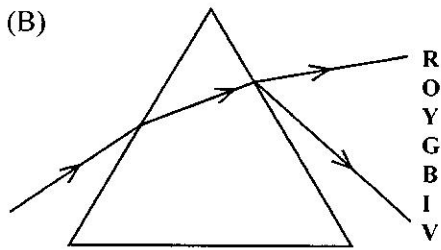
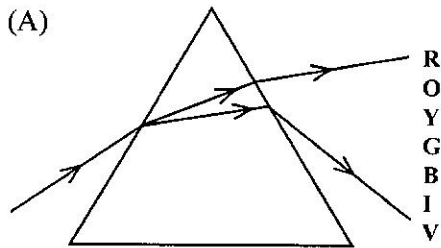
33. Which of the following diagrams illustrates the law of reflection?



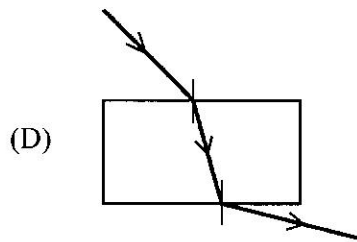
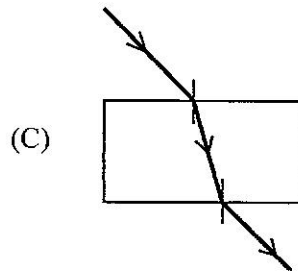
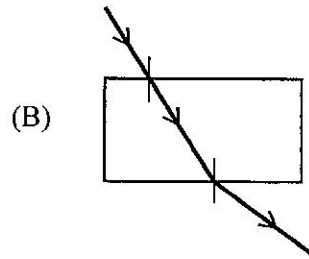
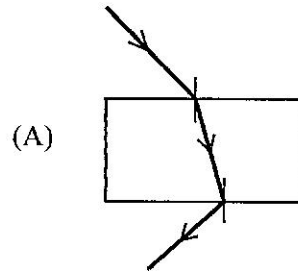
34. Which of the following types of electromagnetic radiation causes tanning?

- (A) Light
- (B) X-ray
- (C) Infrared
- (D) Ultraviolet

35. A ray of white light enters a transparent glass prism. In which of the following diagrams is the dispersion of light correctly illustrated?



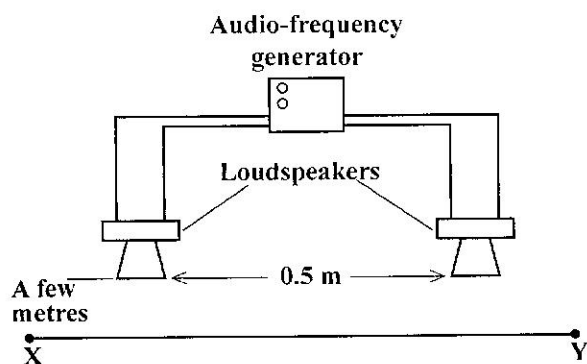
36. Which diagram BEST shows the path taken by a ray of light through a rectangular block?



37. A ray of light leaves the air and enters glass of refractive index 1.5. If the angle of incidence is 37° , what is the sine of the angle of refraction?

(A) $1.5 \sin 37^\circ$
 (B) $1.5 + \sin 37^\circ$
 (C) $\frac{1.5}{\sin 37^\circ}$
 (D) $\frac{\sin 37^\circ}{1.5}$

Item 38 refers to the following diagram. Two similar loudspeakers are connected to the same audio-frequency generator and set up as shown.



38. At some points along XY, no sound is heard because
- (A) the sound waves are diffracted
 (B) sound is refracted away from those points
 (C) interference of the sound waves takes place
 (D) the sound waves are reflected back to the same source

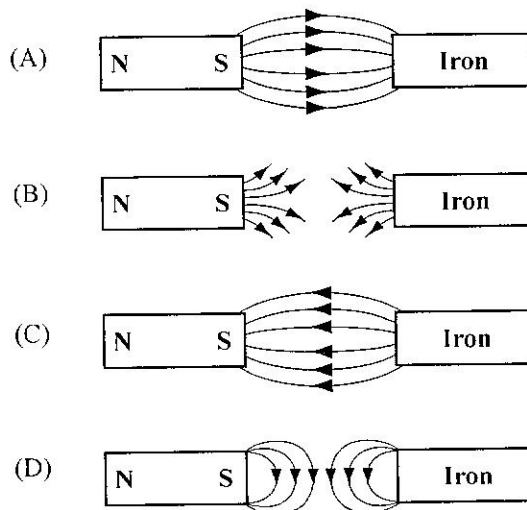
39. The human ear is incapable of hearing a silent dog whistle because

(A) it does not make a noise
 (B) the speed of sound is too fast to be detected by the human ear
 (C) the frequency of the waves are above the range detected by the human ear
 (D) the waves are infrasound with frequencies below that detectable by the human ear

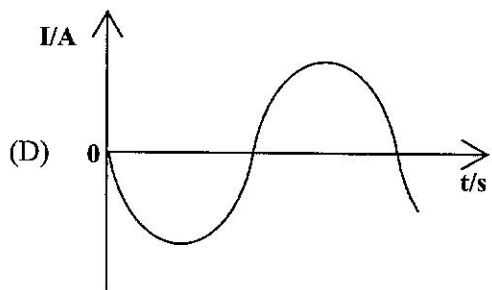
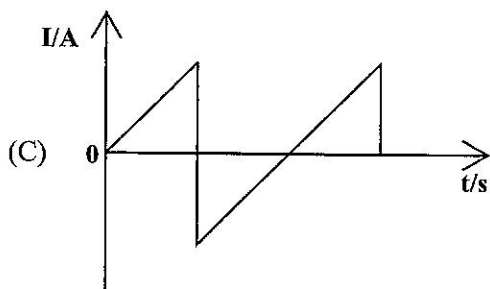
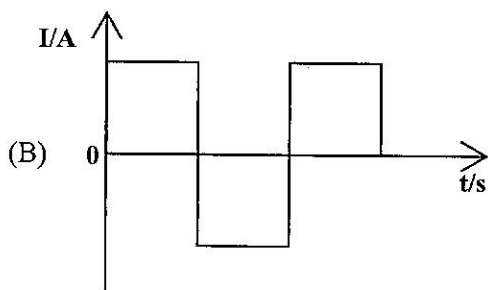
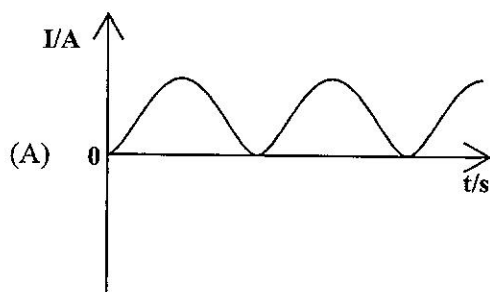
40. Which of the following electromagnetic waves has the SHORTEST wavelength?

(A) Gamma rays
 (B) Infrared waves
 (C) Radio waves
 (D) Ultraviolet radiation

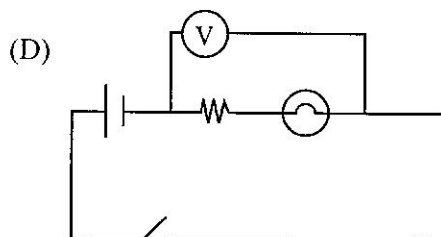
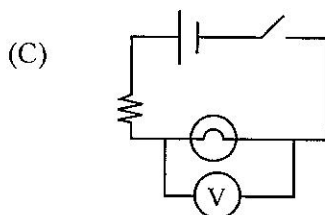
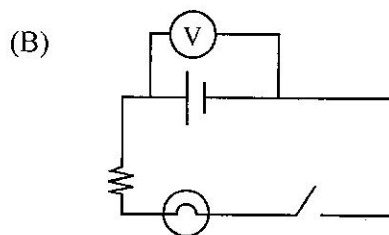
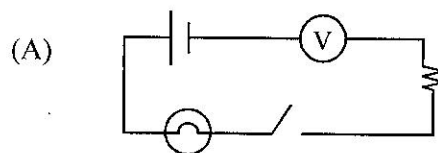
41. Which of the following diagrams shows the magnetic field formed between a bar magnet and a piece of iron?



42. Which of the following current (I)–time (t) graphs BEST represents a d.c. current?



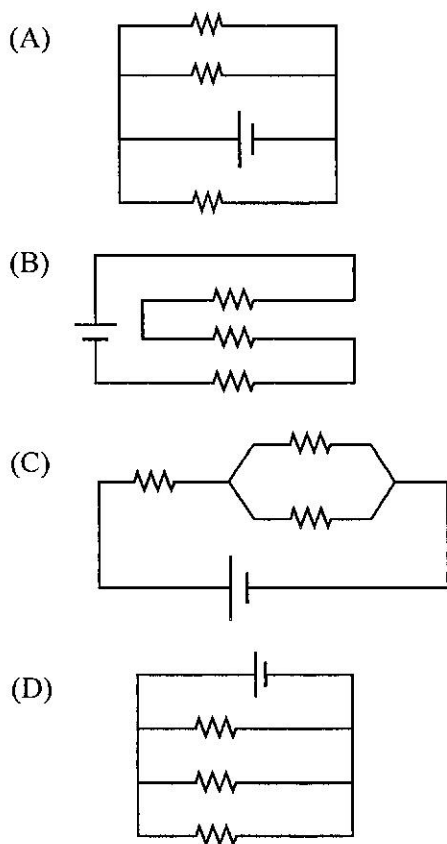
43. Which of the following circuit arrangements is BEST suited for measuring a lamp's voltage?



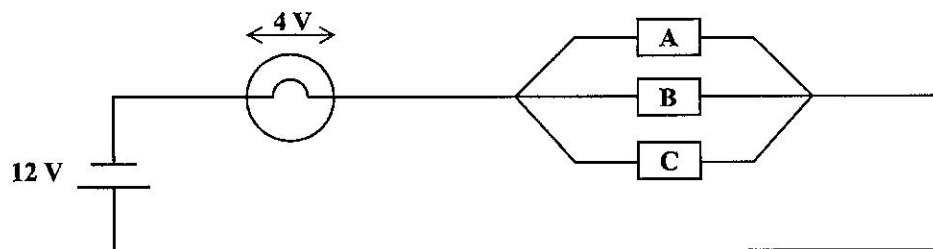
44. The unit of potential difference is defined as one

- (A) joule per volt
- (B) volt per ampere
- (C) joule per ampere
- (D) joule per coulomb

45. Which of the following circuit diagrams BEST represents a series arrangement?

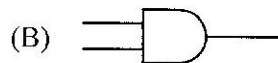


Item 46 refers to the following diagram



46. Given that the potential difference across the lamp in the circuit is 4 V, what is the potential difference across resistor A?
- (A) $(12 \div 4) \text{ V}$
 (B) $(12 \times 4) \text{ V}$
 (C) $(12 - 4) \text{ V}$
 (D) $(12 + 4) \text{ V}$

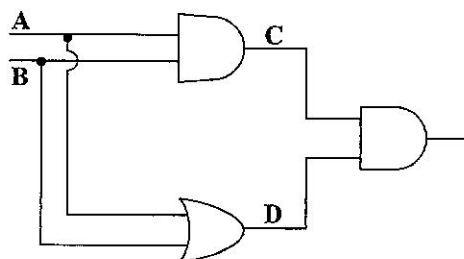
47. Which symbol represents the NAND gate?



49. An ideal transformer has a primary to secondary turns ratio of 1:3. An alternating potential difference of 200 V is applied to the primary coil and a resistance of $200\ \Omega$ attached to the secondary coil. What is the current in the secondary circuit?

- (A) 0.33 A
(B) 1.00 A
(C) 1.50 A
(D) 3.00 A

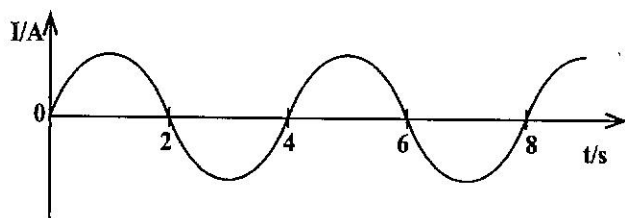
Item 48 refers to the following diagram.



48. What is the output at C and D when the input at A is 0 and input at B is 1.

	C	D
(A)	0	0
(B)	0	1
(C)	1	0
(D)	1	1

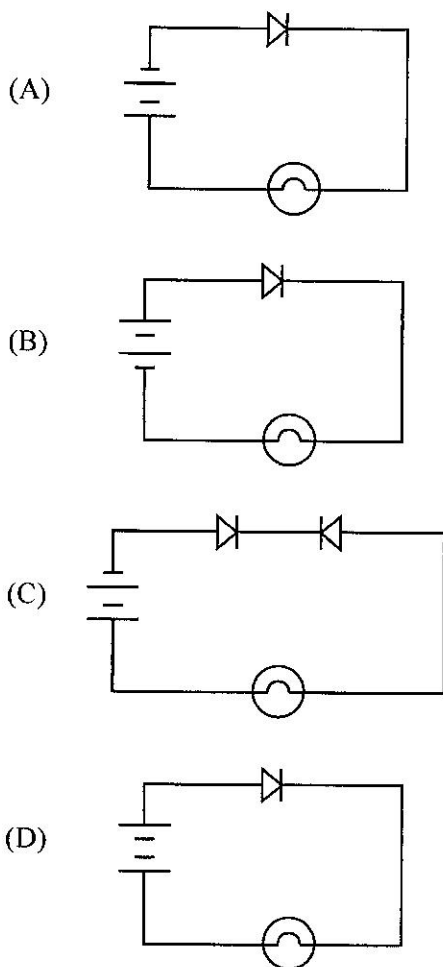
Item 50 refers to the following current–time graph.



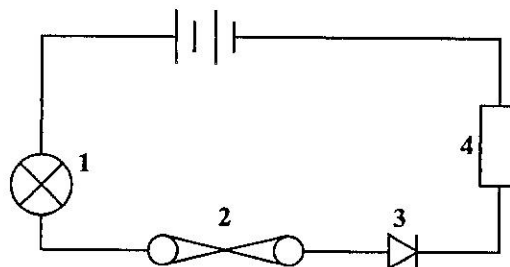
50. What is the period of the current?

- (A) 2 s
- (B) 4 s
- (C) 6 s
- (D) 8 s

51. In which of the following circuits will the lamp light up?



Item 52 refers to the following diagram.

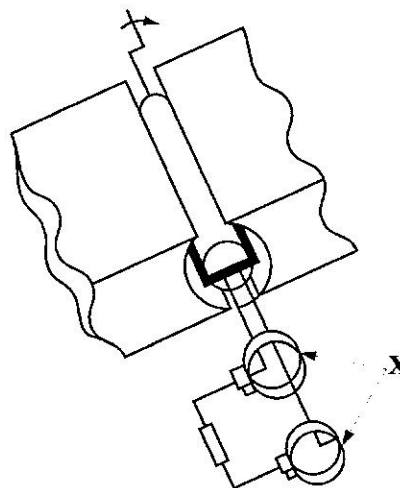


52. Which of the following options shows the correct match of the component name and its number?

Diode Lamp Resistor Fuse

- | | | | | |
|-----|---|---|---|---|
| (A) | 3 | 1 | 4 | 2 |
| (B) | 4 | 3 | 1 | 2 |
| (C) | 1 | 3 | 4 | 2 |
| (D) | 2 | 4 | 3 | 1 |

Item 53 refers to the following diagram of a simple a.c. generator.



53. The parts labelled X in the diagram are known as the

- (A) coils
- (B) armatures
- (C) commutators
- (D) slip rings

54. Half-life is defined as
- (A) the time it takes half of the radioactive atoms in a sample to decay
 - (B) the time it takes the radioactive atoms in a sample to decay
 - (C) half the time it takes a radioactive nucleus to decay
 - (D) half the lifetime of a radioactive sample

55. Which of the following scientists discovered the relationship $E = mc^2$?

- (A) Marie Curie
- (B) Isaac Newton
- (C) J J Thompson
- (D) Albert Einstein

56. Which of the following CANNOT be deflected by a magnetic field?

- (A) Alpha particles
- (B) Beta particles
- (C) Gamma rays
- (D) Electrons

57. The nuclide ${}_{90}^{234}\text{Th}$ contains

- (A) 90 protons and 234 neutrons
- (B) 235 protons and 90 neutrons
- (C) 90 protons and 144 neutrons
- (D) 144 protons and 90 neutrons

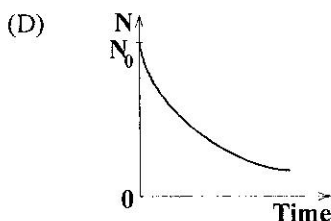
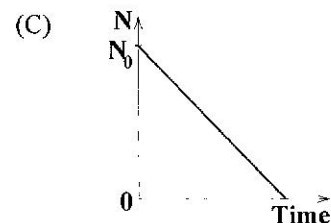
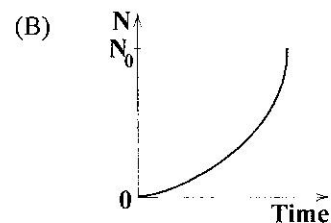
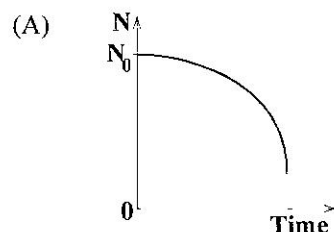
58. Which of the following statements about radioactive decay are correct?

- I. It is a random process.
- II. It is dependent on conditions external to the nucleus.
- III. It is due to changes in the nuclei of atoms.

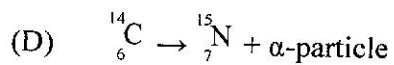
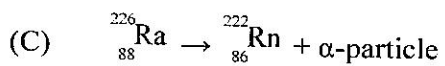
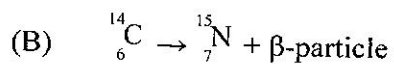
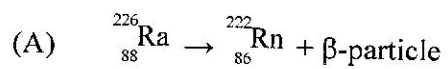
- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

59. N_0 radioactive nuclei are present in a sample at time $t = 0$.

Which of the following graphs BEST represents the variation with time of the number, N , of undecayed nuclei present?



60. Which of the following equations is correct?



END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.